



(Year 2 / A2 Level)

Thermodynamics Lecture Notes

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CHAPTER 16

Thermodynamics

16.1 Internal energy

we now consider the total energy within a thermodynamic system

molecules in a system undergo random motion, so they have kinetic energy

there are potential energy between molecules due to intermolecular interaction

internal energy is defined as the sum of random kinetic energy of molecules and potential energy between molecules: $U = E_k + E_p$

➤ internal energy is a *state function* of the system

it only depends on current state of system, not on process to arrive at this state

16.1.1 Kinetic energy

➤ internal energy counts K.E. due to random motion at molecular level

K.E. of macroscopic motion of the system as a whole is not included

➤ mean K.E. of molecules is directly proportional to temperature: $E_k \propto T$

K.E. of molecules depends on temperature only

higher temperature means molecules move faster, vibrate more intensively, etc.

16.1.2 Potential energy

➤ internal energy counts P.E. due to force fields *within* the system

P.E. of the system as a whole due to *external* force fields is not included

➤ P.E. between molecules depends on intermolecular separation and chemical bonding

in general, greater intermolecular separation means greater P.E. (e.g., melting ice: separation decreases but H-bonds break, so P.E. still increases): $r \uparrow \Leftarrow E_p \uparrow$

breaking intermolecular bonds also causes an increase in P.E.

mean P.E. of gas > mean P.E. of liquid > mean P.E. of liquid solid

16.1.3 Internal energy of ideal gas

for ideal gas, mean K.E. of one molecule: $E_k = \frac{3}{2}kT$

there is no intermolecular force, so P.E. of ideal gas is defined to be zero: $E_p = 0$

internal energy per molecule: $U = E_k + E_p = \frac{3}{2}kT$

hence internal energy of ideal gas is purely kinetic and directly proportional to temperature

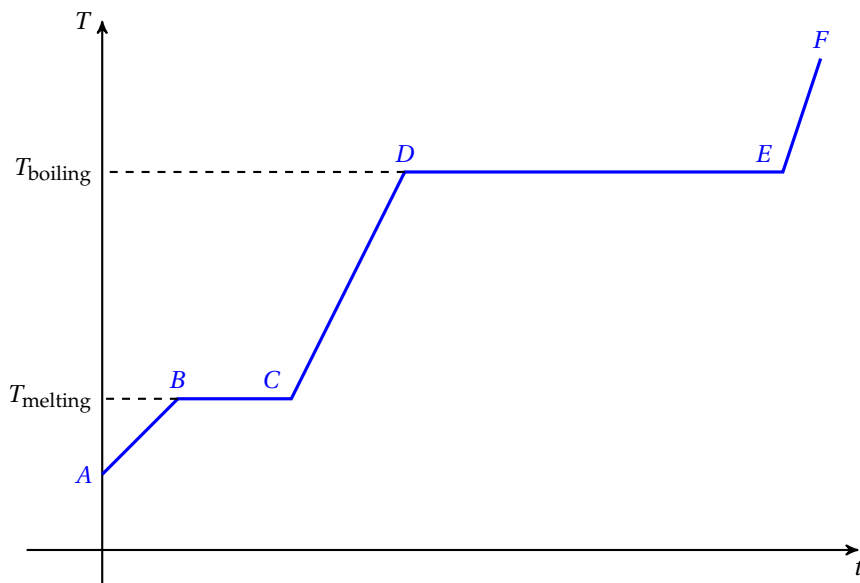
total internal energy of the gas: $U_{\text{gas}} = NU \Rightarrow U_{\text{gas}} = \frac{3}{2}NkT$

16.1.4 Change of states

consider a substance being heated from its solid state

it will melt into a liquid and further vaporise into its gaseous state

we now look into the changes of internal energy during each stage



- *AB (solid state)*: $T \nearrow \Rightarrow E_k \nearrow$, greater vibration for solid particles
but no (significant) change in mean separation (solids expand slightly when heated, so separation increases a little) $\Rightarrow$ no change in P.E.
- *BC (melting)*: latent heat goes into breaking intermolecular bonds, $r \nearrow \Rightarrow E_p \nearrow$
but melting occurs at constant temperature (pure substances melt at a fixed melting point; mixtures melt over a range) $\Rightarrow$ no change in K.E.
- *CD (liquid state)*: $T \nearrow \Rightarrow E_k \nearrow$, greater vibration and free motion
but no (significant) change in mean separation $\Rightarrow$ no change in P.E.
- *DE (boiling)*: molecules break free, $r \nearrow \Rightarrow E_p \nearrow$
boiling occurs at constant temperature (pure substances boil at a fixed boiling point; mixtures boil over a range) $\Rightarrow$ no change in K.E.
- *EF (gas state)*: $T \nearrow \Rightarrow E_k \nearrow$ particles move even faster
particles completely separated, no intermolecular force, so constant $E_p = 0$

Question 16.1 For a particular substance, why is the specific latent heat of vaporisation much greater than the specific latent heat of fusion?

Evaporation

liquid changes into gas without boiling $\rightarrow$ **evaporation**

particles move randomly, i.e., they move at various speeds

some molecules move fast enough to break free

➤ *cooling effect*: evaporation causes a decrease in temperature of the liquid

most energetic molecules escaped, those remain in the liquid have less energy, $E_k \searrow \Rightarrow T \searrow$

➤ rate of evaporation increases with temperature, surface area of liquid

➤ different between boiling and evaporation

	boiling	evaporation
occurrence	throughout the liquid	at surface only
temperature	occur at boiling point	occur at any temperature
bubble formation	bubbles formed	no bubbles
rate of process	fast	slow

16.1.5 First law of thermodynamics

internal energy of a system changes upon heat transfer or doing work

first law of thermodynamics states that the increase in internal energy equals sum of heat supply to the system and work done on the system: $\Delta U = Q + W$

- > first law of thermodynamics is an extension of the law of conservation of energy
 - > sign conventions for Q and W
 - $Q > 0$ if heat supplied to system
 - $Q < 0$ if heat released by system to surroundings
 - $W > 0$ if work done *on* system by external
 - i.e., if system is compressed and volume decreases, then $W > 0$
 - $W < 0$ if system does work *against* surroundings
 - i.e., system expands and volume increases, then $W < 0$
 - > amount of heat energy: $Q = \begin{cases} cm\Delta T & \text{(if no change of state)} \\ Lm & \text{(during change of state)} \end{cases}$
 - > amount of work is related to pressure and change of volume
 - if volume changed at *constant* pressure, then $W = F\Delta s = pA\Delta s \Rightarrow W = p\Delta V$ (if pressure varies, $W = \int p dV$, or find area under p - V graph)
 - if no change of volume, then no work is done
- Example 16.1** A gas is heated by supplying it with 25 kJ of energy. The gas expands so that the volume increases by 0.10 m³. Assume the gas has a fixed pressure of 150 kPa during the process. Calculate the change in internal energy.

amount of work done: $W = p\Delta V = 150 \times 0.10 = 15 \text{ kJ}$

but gas expands means work is done against surroundings, so this is negative work

change in internal energy: $\Delta U = Q + W = (+25) + (-15) = +10 \text{ kJ}$

□

Example 16.2 Use the idea of internal energy and the first law of thermodynamics, explain why boiling water requires heat supply.

boiling occurs at constant temperature, so $\Delta E_k = 0$

but separation between molecules increased, so $\Delta E_p > 0$

by definition, internal energy $U = E_k + E_p$, so $\Delta U > 0$

during boiling, there is an increase in volume, so work against surroundings, $W < 0$

recall first law of thermodynamics $\Delta U = Q + W$, must have $Q > 0$

this means heat must be supplied for boiling processes

□

Example 16.3 When you pump up a bicycle tyre, the temperature of air inside the tyre will go up. Explain why this happens using the first law of thermodynamics.

pumping up tyre involves compressing gas, so positive work is done: $W > 0$

for each stroke, there is little time for heat transfer, so $Q \approx 0$

according to first law of thermodynamics $\Delta U = Q + W \Rightarrow \Delta U > 0$

by definition, internal energy $U = E_k + E_p \Rightarrow \Delta U = \Delta E_k + \Delta E_p > 0$

but for gas, there is negligible intermolecular force, so $\Delta E_p = 0$, then must have $\Delta E_k > 0$

K.E. of molecules is proportional to temperature, higher K.E. so higher temperature

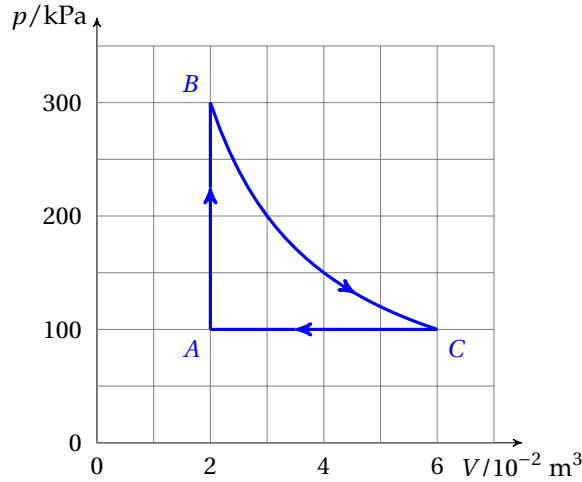
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Example 16.4 An ideal gas of 0.080 mol is initially at state A and then undergoes a cycle $ABCA$.

The variation of its pressure p with its volume V is shown on the graph.

Temperature of state A is 300 K. The magnitude of work on gas from state B to C is 6570 J.

For each stage $A \rightarrow B$, $B \rightarrow C$ and $C \rightarrow A$ during the cycle, determine work done and heat supply to the gas, and also find the change in internal energy.



 work done depends on change in volume

$A \rightarrow B$: no change in volume, so $W_{AB} = 0$

$B \rightarrow C$: $|W_{BC}| = 6570 \text{ J}$, but expansion implies $W < 0$, so $W_{BC} = -6570 \text{ J}$

$C \rightarrow A$: $|W_{CA}| = p\Delta V_{CA} = 1 \times 10^5 \times (6 - 2) \times 10^{-2} \Rightarrow W_{CA} = +4000 \text{ J}$ (compression so $W > 0$)

change in internal energy of ideal gas depends on change in temperature

$A \rightarrow B$: same V but $p_B = 3p_A$, so $T_B = 3T_A = 900 \text{ K}$

$$\Delta U_{AB} = \frac{3}{2} Nk\Delta T_{AB} = \frac{3}{2} \times 0.80 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (900 - 300) \approx +5980 \text{ J}$$

$B \rightarrow C$: note that $p_B V_B = p_C V_C$, so $T_B = T_C$, no change in temperature, so $\Delta U_{BC} = 0$

$$C \rightarrow A: \Delta U_{CA} = \frac{3}{2} Nk\Delta T_{CA} = \frac{3}{2} \times 0.80 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (300 - 900) \approx -5980 \text{ J}$$

for cycle $ABCA$, same initial and final state, so total change in internal energy must be zero

$$\text{one can check that } \Delta U_{\text{cycle}} = \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$$

to find supply of thermal energy, we apply first law of thermodynamics: $\Delta U = Q + W$

$$A \rightarrow B: +5980 = Q_{AB} + 0 \Rightarrow Q_{AB} = +5980 \text{ J}$$

$$B \rightarrow C: 0 = Q_{BC} + (-6570) \Rightarrow Q_{BC} = +6570 \text{ J}$$

$$C \rightarrow A: -5980 = Q_{CA} + (+4000) \Rightarrow Q_{CA} = -9980 \text{ J}$$

the table below summarises all energy changes during the cycle $ABCA$

change	W/J	Q/J	$\Delta U/\text{J}$
$A \rightarrow B$	0	+5980	+5980
$B \rightarrow C$	-6570	+6570	0
$C \rightarrow A$	+4000	-9980	-5980

□

Question 16.2 Show that when n mol of gas is heated at a fixed volume, thermal energy required to raise the temperature by 1.0 K is nR .

Question 16.3 Two identical balloons A and B hold the same amount of gas at the same initial temperature. They are given the same amount of heat. Suppose volume of A is fixed, while B is allowed to expand, compare the final temperatures of the gases in the two balloons.